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ST ANDREW'S JUNIOR COLLEGE



JC2 Preliminary Examinations

Chemistry

8873/1

Higher 1

17 Sep 2018

Paper 1

1300 – 1400

Candidates answer on separate paper.

Additional Materials: Data Booklet and OAS

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, civics group and index number on the OAS provided unless this has been done for you.

There are **thirty** questions on this paper. Answer all questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the one you consider correct and record your choice in **soft pencil** on the separate OAS.

Read the instructions on the OAS very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

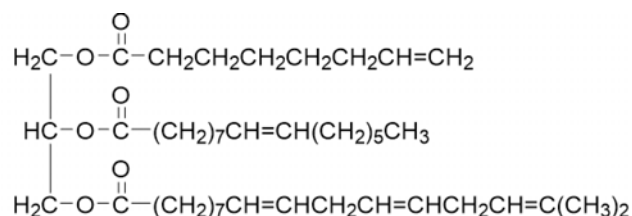
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- 1 Use of the Data Booklet is relevant to this question.

Which sample of gas contains the same number of atoms as 2 g of hydrogen gas?

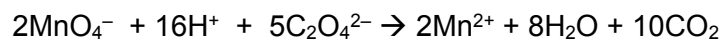
- A 22 g of carbon dioxide
 B 12 g of steam
 C 10 g of neon
 D 8 g of methane

- 2 The diagram shows the structure of a tri-ester.



What is the total number of cis-trans isomers possible for this tri-ester?

- A 2^1 B 2^3 C 2^4 D 2^5
- 3 Ethanedioate ions, $\text{C}_2\text{O}_4^{2-}$, are oxidised by acidified aqueous potassium manganate(VII) to give carbon dioxide according to the following equation:

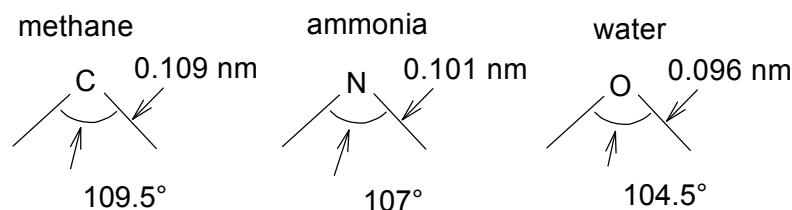


What volume of $0.020 \text{ mol dm}^{-3}$ potassium manganate(VII) is required to completely oxidise $1.0 \times 10^{-3} \text{ mol}$ of the salt $\text{KHC}_2\text{O}_4 \cdot \text{H}_2\text{C}_2\text{O}_4$?

- A 20 cm^3 B 40 cm^3
 C 80 cm^3 D 100 cm^3

[Turn Over

- 4 The bond lengths and bond angles in the molecules of methane, ammonia and water may be represented as follow:



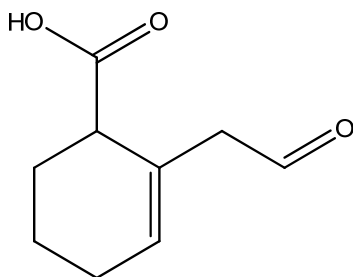
What causes this trend in the bond angles shown?

- 1 Lone pair–bond pair repulsion is greater than bond pair–bond pair repulsion.
 - 2 Decreasing repulsion between the bond pair of electrons as the electronegativity of the central atom increases.
 - 3 Increasing repulsion between hydrogen atoms as the bond length decreases.
- A** 1 only
B 1 and 2 only
C 2 and 3 only
D 1, 2 and 3
- 5 2,3-Dimethylbutane can undergo free radical substitution with chlorine gas. Ignoring any stereoisomers, how many different mono-chlorinated organic products are formed when it is heated with limited chlorine gas?
- A** 2 **B** 3 **C** 4 **D** 6
- 6 Which statement about Group 1 metals is correct?
- A** Melting point increases down the group.
B Reducing strength increases down the group.
C Each ground state atom contains a pair of electrons in its valence shell.
D The second electron is more difficult to be removed than the first electron, as M^+ has a noble gas configuration.

[Turn Over

- 7** An enzyme may be described as being specific in its activity.
What is meant by this statement?
- A** The enzyme will denature at high temperature.
 - B** The enzyme helps to increase the speed of a chemical reaction.
 - C** The enzyme will catalyse a narrow range of chemical reactions.
 - D** The enzyme is a biological catalyst which remains chemically unchanged at the end of the reaction.
- 8** Chlorate (I) ion undergoes disproportionation to give chloride ion and another chlorine containing product.
Which of the following could be the chlorine-containing product?
- A** Cl_2
 - B** Cl_2O
 - C** ClO^-
 - D** ClO_3^-

- 9 Which are the products formed when compound **X** reacts with the following reducing agents?



compound **X**

	H_2 , Pt	LiAlH_4 in dry ether
A		
B		
C		
D		

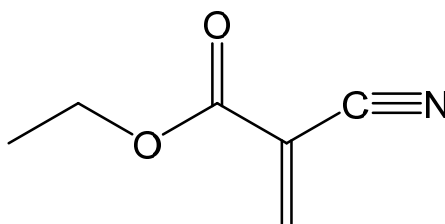
10 A substance **Y** has the following properties.

- melting point – 39 °C
- boiling point 357 °C
- conduct electricity when solid and molten

What is the structure of **Y**?

- A** ionic
- B** metallic
- C** giant molecular
- D** simple molecular

11 Ethyl cyanoacrylate, $C_6H_7O_2N$, is the major component of superglue.



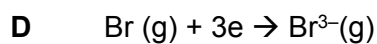
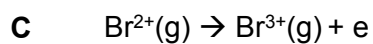
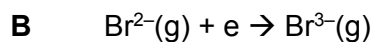
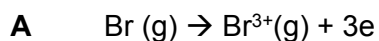
ethyl cyanoacrylate

Which statements about the molecule are true?

- 1 It contains 5 lone pairs of electrons.
- 2 It reacts with Br_2 to give a pair of cis-trans isomers.
- 3 It contains 4 π bonds.
- 4 It can be hydrolysed by dilute NaOH to give a carboxylic acid and an alcohol.

- A** 1 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 3 and 4

12 Which equation represents the third ionisation energy of bromine?



13 When 20 cm^3 of $0.5 \text{ mol dm}^{-3} \text{ NaOH(aq)}$ is neutralised with an equal volume of $0.5 \text{ mol dm}^{-3} \text{ HNO}_3(\text{aq})$, the temperature of the mixture rose by 5.2°C .

What would be the temperature change if 40 cm^3 of $0.25 \text{ mol dm}^{-3} \text{ NaOH(aq)}$ is reacted with an equal volume of $0.25 \text{ mol dm}^{-3} \text{ HNO}_3(\text{aq})$?

A 1.3°C

B 2.6°C

C 5.2°C

D 10.4°C

14 Iodine-131 is a radioactive isotope with a half-life of 10 days. Given that radioactive decay is a first-order reaction, what fraction of the isotope would remain after 50 days?

A $\frac{1}{10}$

B $\frac{1}{20}$

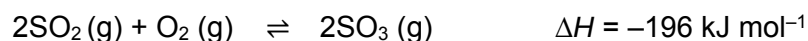
C $\frac{1}{32}$

D $\frac{1}{50}$

- 15** The molar mass of ethanoic acid is 120.0 g mol^{-1} when dissolved in ether but 60.0 g mol^{-1} when dissolved in water.

Which one of the following statements best explains the phenomenon?

- A** Ethanoic acid is more soluble in ether than in water.
 - B** Ethanoic acid molecules form strong covalent bonds with ether molecules.
 - C** Ethanoic acid molecules form extensive hydrogen bonds with water molecules.
 - D** Ethanoic acid molecules form dative bonds with another ethanoic acid molecule when dissolved in ether.
- 16** The enthalpy change of reaction for the conversion of sulfur dioxide to sulfur trioxide is shown below.



Which of the following statements is correct?

- A** When the volume of the vessel is compressed, the yield of SO_3 decreases.
- B** When a catalyst is introduced to the system, the yield of SO_3 increases.
- C** When the temperature is increased, only the rate of the backward reaction is increased.
- D** When the temperature is increased, K_c decreases.

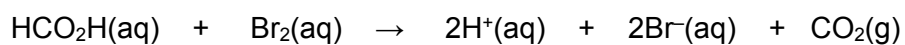
- 17** Bromine has two isotopes, ^{79}Br and ^{81}Br .

Propene can react with bromine liquid in the dark to form organic products with different relative molecular mass.

How many different relative molecular mass values does the organic products have?

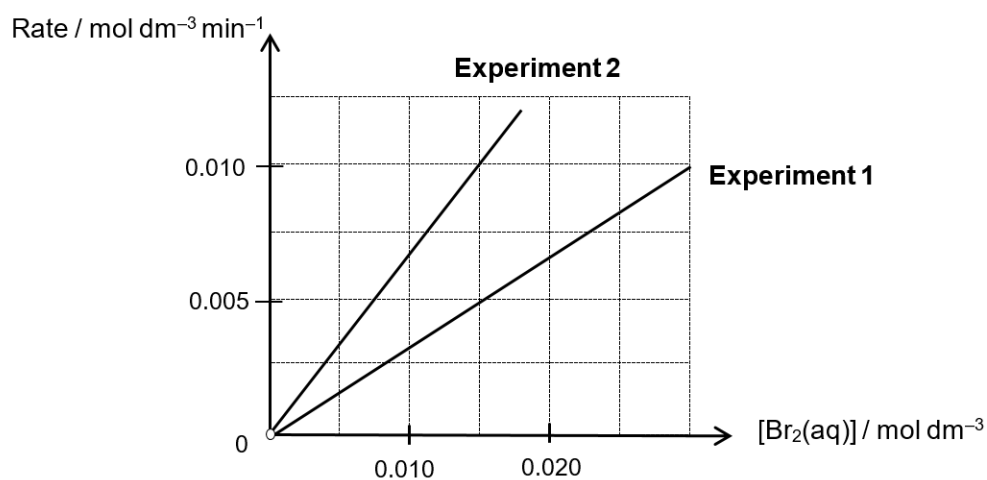
- A** 2
- B** 3
- C** 4
- D** 5

- 18 Two separate experiments were conducted to study the kinetics of the following reaction in the presence of excess methanoic acid.



Experiment	Initial $[\text{HCO}_2\text{H}] / \text{mol dm}^{-3}$
1	0.8
2	1.6

The experimental results were represented in the graph as shown.



Which of the following is the correct rate equation for above the reaction?

- A Rate = $k [\text{Br}_2]$
- B Rate = $k [\text{HCO}_2\text{H}]$
- C Rate = $k [\text{HCO}_2\text{H}][\text{Br}_2]$
- D Rate = $k [\text{HCO}_2\text{H}]^2[\text{Br}_2]$

- 19 The chloride of **X** and oxide of **Y** respectively are separately mixed with water. The two resulting mixtures have different effect on litmus.

Which of the following represents the correct elements?

	X	Y
A	Al	S
B	Mg	P
C	Si	P
D	P	Na

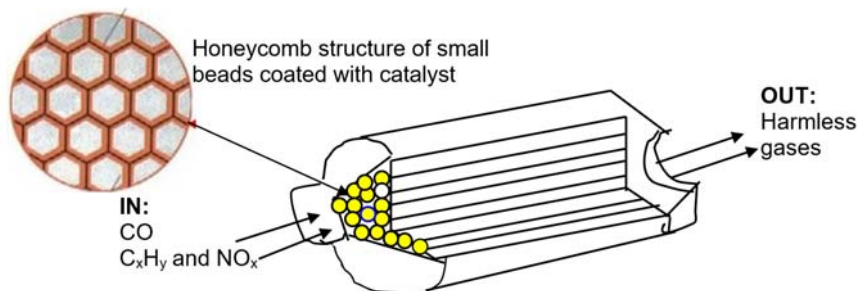
- 20 A pure sample of $\text{NH}_3(\text{g})$ is introduced into a vessel of constant volume and an equilibrium is established as shown below.



The value of the final pressure is then found to be 25 % greater than if only NH_3 were present. What is the mole fraction of N_2 in the reaction mixture?

- A** 0.10
B 0.25
C 0.45
D 0.75
- 21 What is the correct description of the bonding between the two carbon atoms in benzene?
- A** two pi bonds that are longer than a C—C single bond.
B two pi bonds that are shorter than a C—C single bond.
C a sigma and a pi bond that are longer than a C—C single bond.
D a sigma and a pi bond that are shorter than a C—C single bond.

- 22** Catalytic converters in car exhaust system helps to convert harmful emission gases into harmless ones. The following diagram shows the interior of a catalytic converter.



Which of the following statement is correct?

- A** The forces of attraction holding the carbon monoxide, unburnt hydrocarbons and oxides of nitrogen onto the surface of the catalyst are stronger than the forces of attraction holding on the product molecule onto the surface.
 - B** The catalyst coated on the honeycomb surface provides a small surface area to volume ratio to increase the rate of reaction.
 - C** Homogenous catalyst is used in the catalytic converted to lower the activation energy for conversion of harmful gases
 - D** Nitrogen dioxide gas is one of the products of the reactions.
- 23** What is the electronic configuration of copper in Cu₂O?

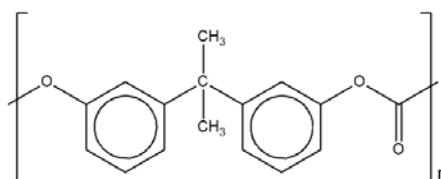
- A** [Ar] 3d⁹
- B** [Ar] 3d¹⁰
- C** [Ar] 3d⁹ 4s¹
- D** [Ar] 3d⁹ 4s²

24 Which of the following statements about the Period 3 elements sodium to chlorine, and their compounds, is correct?

- 1** The ionic radius decreases across the period.
- 2** The oxide of the elements become more acidic across the period.
- 3** The boiling points of the elements decreases in the order phosphorus, sulfur, chlorine.

- | | |
|-----------------------|-----------------------|
| A 2 only | B 2 and 3 only |
| C 1 and 2 only | D 1,2 and 3 |

25 Polycarbonates (PC) are polymers containing carbonate groups in their chemical structures. Some major smartphone manufacturers such as Samsung and Apple started using PC in their phones casing.



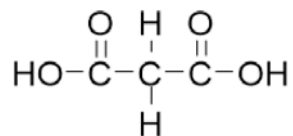
PC

Which row correctly describes PC?

	type of polymer	reaction of polymer with ethanoic acid
A	addition	no reaction
B	condensation	no reaction
C	addition	hydrolysis
D	condensation	hydrolysis

- 26** Which reaction will give propanoic acid as one of the products?
- A** heating $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ under immediate distillation with potassium dichromate.
- B** heating $\text{CH}_3\text{CH}_2\text{CONHCH}_3$ under reflux with dilute aqueous sulfuric acid.
- C** heating $\text{CH}_3\text{CH}_2\text{OCOCH}_3$ under reflux with dilute aqueous sulfuric acid.
- D** heating $\text{CH}_3(\text{CH}_2)_2\text{COOCH}_3$ under reflux with dilute aqueous sulfuric acid.
- 27** What is the maximum size, in at least one dimension, of a nanomaterial?
- A** $1 \times 10^{-6} \text{ m}$
- B** $1 \times 10^{-7} \text{ m}$
- C** $1 \times 10^{-8} \text{ m}$
- D** $1 \times 10^{-9} \text{ m}$
- 28** The two most common isotopes of titanium are $^{56}_{22}\text{Ti}$ and $^{58}_{22}\text{Ti}$. Which statement about the isotopes of titanium is correct?
- A** Both isotopes have same numbers of neutrons.
- B** One of the isotopes has more protons than the other.
- C** The electronic configuration of Ti^{2+} ion for both isotopes is the same.
- D** In the same electric field strength, $^{58}_{22}\text{Ti}^+$ will be deflected more than $^{56}_{22}\text{Ti}^+$.

- 29** The diagram shows the structure of the malonic acid.



Which of the following shows the correct set of the bond angles present in the malonic acid?

- A** 105° , 109° , 120°
- B** 109° , 180° only
- C** 120° , 180° only
- D** 105° , 109° only
- 30** Ammonia gas can react with sodium hydride in the following equation:



Which statement correctly describes the behaviour of ammonia in this reaction?

- A** It acts as an Arrhenius acid.
- B** It acts as an Arrhenius base.
- C** It acts as Bronsted-Lowry acid.
- D** It acts as Bronsted-Lowry base.

END OF PAPER